

ICS 4111: Embedded Systems & IoT Practical

Lab: Using Micro python For IoT

Instructions

In this lab you will develop a complete IoT device/Node Using ESP32 microcontroller, DHT22 to monitor temperature and humidity, and then transmit the data over WiFi using the MQTT messaging protocol and store it in the database.

- A. You will use Micro python as programming language
- B. Once done, publish your work to your GitHub and share the link through e-learning together with the documentation

1. Objectives

By the end of this lab you will be able to:

- ✓ Wire a DHT22 sensor correctly to an ESP32 development board
- ✓ Flash MicroPython firmware onto the ESP32 using esptool
- ✓ Write MicroPython code to read sensor data and connect to WiFi
- ✓ Publish structured JSON data to an MQTT broker
- ✓ Subscribe to an MQTT topic and persist incoming data to a database
- ✓ Debug a multi-layer IoT system end-to-end

2. Hardware Requirements

Component	Qty	Notes
ESP32 Development Board	1	Any variant - Devkit recommended
DHT22 Temperature/Humidity	1	Also sold as AM2302
Breadboard	1	Half or full size
Jumper Wires (Male-to-Male)	3	For sensor-to-board connections
10 kΩ Resistor	1	Pull-up on the DHT22 data line
USB Data Cable	1	Data capable - charge-only cables will not work

3. Software Requirements

Install and verify every software item below on your pc.

Software	Version / Install	Purpose
Thonny IDE	Latest stable	Recommended for MicroPython; built-in REPL support
esptool	pip install esptool	Flashes MicroPython firmware to the ESP32
MicroPython for ESP32	v1.24+	Download .bin from micropython.org/download/ESP32_GENERIC
Mosquitto MQTT Broker	Latest	Local broker; or use broker.hivemq.com for testing
Python 3.x	3.9+	For the PC-side subscriber script

paho-mqtt	pip install paho-mqtt	Python MQTT client library
SQLite	Bundled	Ships with Python, no separate install needed

4. MQTT Protocol Overview

MQTT (Message Queuing Telemetry Transport) is a lightweight publish/subscribe protocol designed for constrained devices exactly like our ESP32.

Role	In This Lab	Description
Publisher	ESP32	Sends (publishes) sensor data to a topic
Broker	Mosquitto	Receives and routes all messages by topic
Subscriber	PC Python script	Listens to a topic and processes incoming messages
Topic	iot/lab/sensor	A named channel like an address for the data
Payload	JSON string	The actual sensor data: {"temperature": 24.5, "humidity": 61.2}

5. Deliverables

1. Source code
2. Wiring diagram or clear annotated photograph of your breadboard circuit
3. Screenshot: REPL output showing live sensor readings
4. Screenshot: mosquitto sub terminal receiving at least 10 JSON messages
5. Screenshot: SQLite SELECT query confirming rows stored with timestamps
6. Lab report (max 5 pages): challenges encountered and how you resolved them

When done, upload the results of all your work as one PDF document on e-learning before the set deadline. Only one person should submit on behalf of the group. In the last section of your document, include evidence of you working as a group. This can be a group photo when conducting the experiments or minutes of team discussions. In case you need more time to interact with the components for this Lab on this beyond the scheduled class time, you can visit the School's Makerspace Lab in the Engineering Labs Building to continue prototyping. Schedule a prototyping session through this [link](#) and forward the invite to the instructor (sitotia@strathmore.edu) and lab technologist (jntonjira@strathmore.edu) to facilitate access to the lab.

Please note, no lab access will be granted without an invite email. Ensure to read the lab rules as provided on e-learning